

CHAMPIONS K.C.S.E REVISION

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MARKING SCHEME

PAPER 1,2 &3

BIOLOGY FORM FOUR EXAM 2020

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**CHAMPIONS K.C.S.E REVISION SERIES EXECUTIVE
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REVISION HISTORY AND GEOGRAPHY**

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BIOLOGY PAPER 1

1. a) Study of cells✓
b) Study of fungi✓
2. Helps to develop scientific skills like observation, recording e.t.c.
Helps to arrange information in orderly manner to avoid confusion
Helps to understand evolutionary or phylogenetic relationship
Helps to know features or characteristics of organisms
Helps to identify organisms
Helps to avoid chaos and confusion among scientists

(Mark first three correct answers)
3. a) Sensory neuron / Afferent neuron
- Cell body of the axon ✓ **(Mark 1st correct answer)**
- Short axon
- No dendrites on the cell body
b) A schwann cells✓ **(Reject Schwann cell)**
D node of ranvier✓ **(rej. Nodes of ranvier)**
c) C transmits impulse from receptors/ sense organs to central nervous system / brain/ spinal cord.
4. Diameter of field of view = 3 mm = 3000 μm✓
No of cells = 20 cells
Length of one cell = $\frac{3000 \mu m}{20}$ ✓ = 150 μm✓

(3 marks)
5. Arachnida **(rej arachnidesx)**
-Four pairs of legs✓ **(reject cephalothorax)**
-Lack antennae✓
6. i) Quantity of dry matter of a given type of organisms or dry weight of a given type of organism per unit area
ii) Natural unit made up of biotic and abiotic factors that interact with one another to form a stable self-sustaining system.
7. a) 
- b) - Ensure / impulses travel in one direction✓
- Control learning/ memory / ensure one neuron can make joints with several other neurons✓
8. a) Collenchyma✓
b) Sclerenchyma✓ / xylem **(first correct answer)**
9. a) Site for respiration/ produce energy
b) Inner membrane greatly folded into cristae to increase surface area for attachment of respiratory enzymes / To increase surface area for respiration.
c) lysosomes - destroy worn out cell organelles and pathogens
Packaging, modification and transport glycoproteins, form lysosomes
10. a) A - Epidermis ✓
B - Vascular bundles✓ / xylem and phloem
b) i) Monocotyledonae
ii) Vascular bundles scattered.
11. a) fern
b) A – sorus rej. sori
B - Pinna
C - Adventitious roots
D - rhizome
c) Division pteridophyta

12. In humid conditions, the air contains so much water vapour that the sweat may not evaporate rapidly enough to produce an adequate cooling effect. This may lead to heat accumulation in which the body temperature rises above normal hence the discomfort.
In hot dry weather, the sweat produced evaporates by use of latent heat from the body hence reducing the body temperatures.
13. When is waterlogged the air spaces are occupied by water; this reduces the amount of oxygen in the soil, causing poor root respiration hence poor uptake of mineral salts
14. a) Glucose
b) Glycogen
15. No production of bile to emulsify fats.
16. a) $X^N X^N$
b) $X^n X^n$
c) $X^N X^n$
17. - Head contains a haploid nucleus that fuses with the ovum nucleus during fertilization to form a zygote
- The neck has (a) centrioles for controlling the axial filaments
- Has a tail with axial filament for propulsion
- Has an acrosome that contains lytic enzymes which dissolve the vitelline membrane of the ovum to allow its penetration
- The middle piece is packed with mitochondria that provide energy for propulsion;
18. a) A - Plumule
B - Radical
C - Micropyle
D - Cotyledon
b) - Store food for the embryo
- After germination, it turns green and photosynthesis
- Protects plumule from mechanical damage.
c) Epigeal
19. a) Shall be a universal recipient; can receive blood from any blood group; while a person with blood group 'O' can only receive blood from a person with blood group O/ AB has no antibodies while has no antigen
b) To check for compatibility
To make sure the blood is free from diseases (such as HIV, Hepatitis B, Etc)
20. a) Ability of the body to identify / recognize foreign antigens and develop mechanisms of destroying item or ability of the body to resist infections
b) Natural immunity is inborn /inherited /passed from parents to offspring while acquired immunity is obtained in life through immunization or through infection and production of antibodies
c) Tuberculosis, poliomyelitis, diphtheria whooping cough, measles
21. i) Vibrio cholerae ✓ (rules of binomial nomenclature should be followed)
ii) Entamoeba histolytica ✓
22. The carbon (IV) oxide produced during respiration is utilized during photosynthesis ✓ while the oxygen produced during photosynthesis is used during respiration. ✓
23. a) A - Has umbilical vein to supply foetus with nutrient and umbilical artery to remove waste products lie CO_2
B - Protect the embryo from shock
- Regulate temperature
b) Widened cervix
Head of foetus directly above the cervix
24. Phenotypically acquired characteristics are not genetically transmitted from parents to offspring's; but only those genotypically controlled are transmitted.
25. a) To replace those members of a species that dies hence ensuring continuity of the species/ improves quality.
b) Helps to remove metabolic waste products from the body; which can be toxic/ poisonous
26. The released ovum is viable for 12- 24 hours while the sperm might live in the female reproductive organs for 2-3 days.
27. a) I – condensation
II – hydrolysis
b) Boiling (the test solution) in a dilute acid;

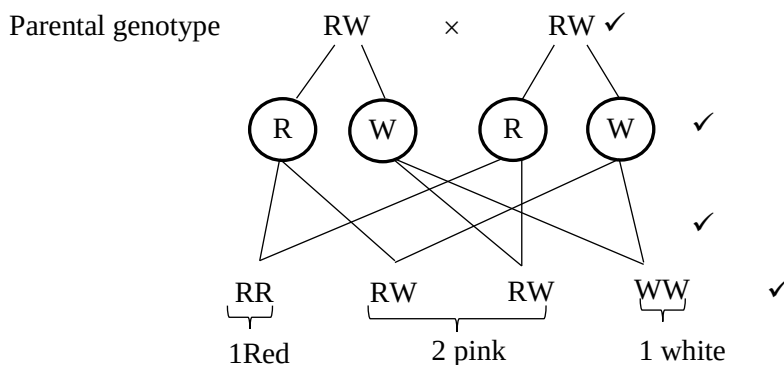
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BIOLOGY PAPER 2

FORM 4

231/2

MARKING SCHEME

1. a) Incomplete dominance ✓
 b) Presence of a different colour from the two there is blending of colours and none of the two Red and White is dominant.
 c) i) Parental phenotype pink × pink



- ii) $\frac{2}{4} \times 1600 = 800 \text{ plants}$ ✓
 $\frac{1}{4} \times 1600 = 400 \text{ plants}$ ✓

2. a) Tertiary consumer / quaternary consumer
 b) i) Land plants → Insects → Lizards → Snakes → Hawks
 ii) Land plants → Insects → Lizards → Hawks
 c) - There will be an increase in the number of birds
 - There will be an increase in the number insects
 - There will be an increase in the number of lizards
 d) Insects
3. a) Positive feedback mechanism
 b) - Excess glucose converted glycogen and stored in liver cells.
 - Glucose oxidized to release energy
 - Some glucose converted to fats and stored in a dispose tissue
 c) i) Pancrease
 Liver
 ii) - To avoid fluctuation in osmotic pressure which can affect function
 - For glucose to be enough for respiration
 d) Maintaining of constant internal osmotic pressure.
4. a) Capture – recapture
 b) - line – transect method
 - belt – transect method
 - Quadrat
 c) $P = \frac{1^{\text{st}} \text{ capture} \times \text{second capture}}{\text{Marked recaptured}}$
 $= \frac{50 \times 90}{25}$ ✓
 $= 180 \text{ fish}$ ✓
 d) - no immigration or emigration of organisms
 - There is free mixing of organisms within the period of study

- The population does not vary within the period of study due to death or birth.
- The markings do not come off or affect the behavior of the organism
- The time period is enough for the organisms to mix uniformly.

5. a) i)

Skeletal muscles

- Are striated
- Has elongated cells
- Multi nucleated

Smooth muscles

- Non striated
- has spindle – shaped cells
- uninucleated

✓ (2x1 = 2marks)

- ii) Biceps / muscle B relax / flex while
Triceps / muscles A contract/ extend the arm stretches.

- b) i) It is made up of several fused sacral vertebrae to form a rigid structure making it firm and strong.
ii) Strong and firm to support body weight and spread it to the legs through the pelvic girdle
iii)
- Fused vertebrae to form rigid structures which are strong and firm to bear body weight.
 - First anterior sacral vertebrae have wing like transverse process fused to the pelvic girdle.
 - Large broad centrum to offer support to body weight
 - Reduced neural spine for muscle attachment
 - Narrow neural canal for passage of spinal cord.

Mark first Two correct answers.

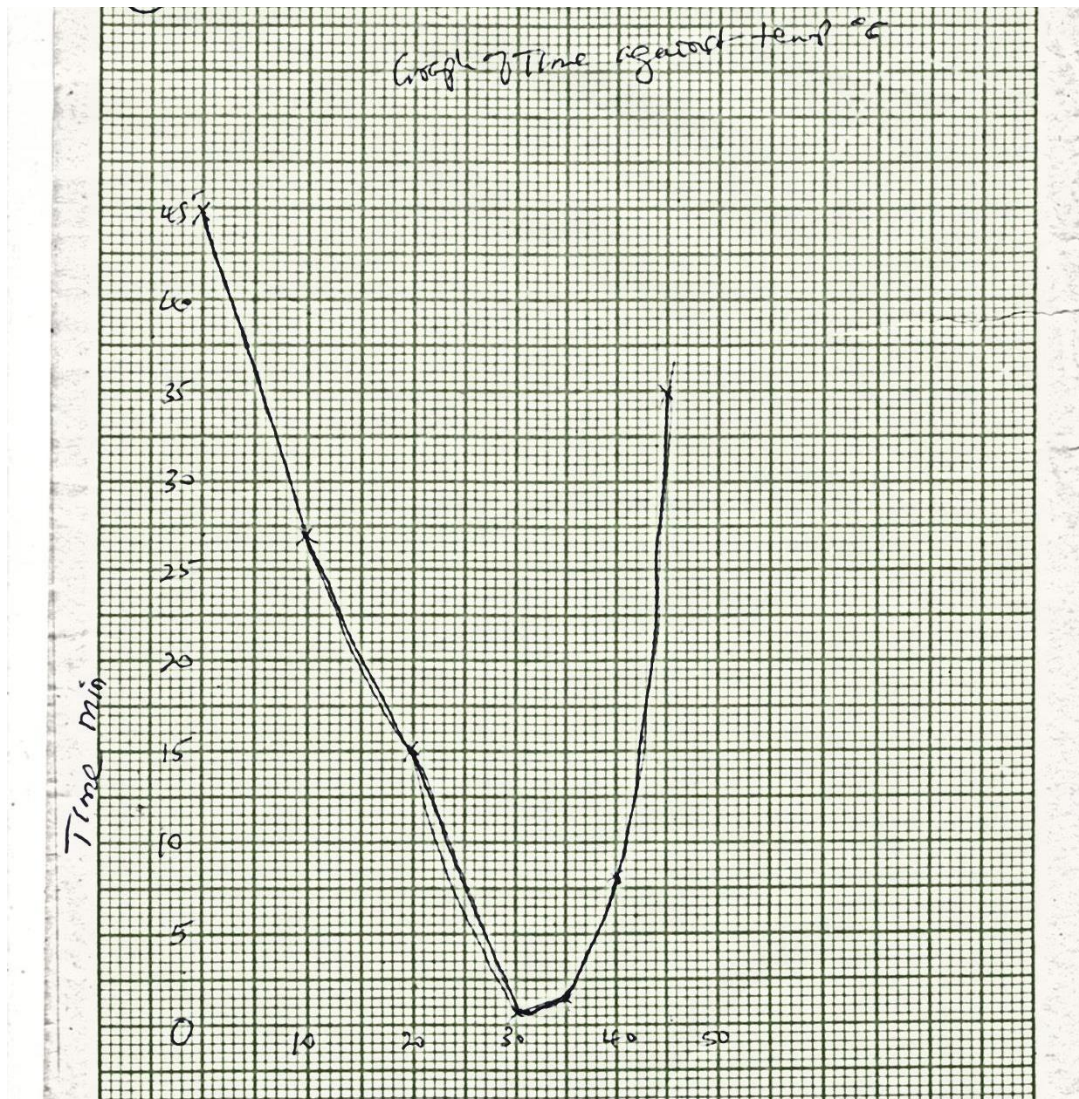
6. a)

Scale on X and Y axis = 2marks

Labeling of axis = 1mark

Plotting of points = 1mark

Curve be drawn using free hand = 1mark



b) $37^{\circ}\text{C} \pm 0.5$

c) i) at 50°C – The time taken is long; temperature too low / way below optimum; enzymes are inactive.

ii) The time taken is long since most enzymes are denatured since temperature is above optimum.

d) Substrate concentration; presence of inhibitors PH range; presence of co-enzyme and co-factors
(award any Two correct answers).

e) Consider the point where a line from 15°C meets the curve, determine the time in minutes divide the time in minutes by the temperature at that point. All the above be shown on the curve

Time in minutes at $15^{\circ}\text{C} = 20.5$

$20.5\text{min} / 15^{\circ}\text{C} = 1.367$ or 1.4

f) Enzyme salivary amylase works in slightly alkaline medium; acidic PH in stomach denatures enzymes

g) Pancreatic Juice

Salivary amylase enzyme

h) HCl neutralizes bicarbonate / pancreatic salts / acid chime from the stomach denatures the pancreatic enzymes.

7.

a) The ear pinna is made up of skin and cartilage and is funnel shaped; it receives and collects sound waves into the auditory meatus; Auditory meatus / ear tube – a canal that is lined with hairs and wax; it allows passage of sound waves to the middle ear; the wax traps dust particles entering into the ear; it also maintains flexibility of the ear drum; the ear drum / tympanic membrane: with a thin tough membrane that covers the external opening of the

middle ear; it vibrates when it is hit by sound waves and transforms sound waves into vibrations; passes vibrations into the ear ossicles: Ear ossicles: has stapes, malleus and incus: three ossicles form a system of levers which amplifies and transmits the vibrations to the oval window; oval window: a thin flexible membrane opening into the cochlea. It receives vibrations from the ossicles and passes them to the inner ear;

The cochlea is a spirally shaped tube filled with endolymph and perilymph; it has sensory cell for hearing; the semicircular canals; consists of three cavities containing endolymph; maintains body balance and posture in relation to movement of the head;

The vestibules – consists of utricle and saccule which contain sensory cells; maintain body balance and posture in relation to gravity;

Eustachian tube; tube connecting the middle ear with the pharynx; equalizes air pressure between the middle ear and the outer ear thus preventing distortion of the eardrum;

Any correct first 10 points.

- b) i)
- Homologous structures e.g. pentadactyl limb in different vertebrates have bones arranged in similar pattern / basic plan but have been modified to perform different functions e.g. digging in moles / grasping in monkey; these indicate organisms have common ancestry;
 - Analogous structures e.g. wings of birds and insects; these structures have different embryonic origin but are modified to perform similar functions;
 - Vestigial structures; some structures become reduced in size over long time and become functionless these are the vestigial structures e.g. caecum and appendix / coccyx in humans; presence of these structures indicates vertebrates have a common ancestry;
- ii) **Geographical distribution**
- Present day continents are thought to have been one large landmass / joined together; this allowed free migration of animals in the land mass;
 - Continental drift led to isolation; led to different patterns of evolution; such that animals occupying the same climatic zones in different continents are different. E.g. Llamas of South America and camels of Africa; lions of Africa and Tiger of Asia / marsupials are unique to Australia;
- 8.
- a) Pollution is the release of chemical/substance / forms of energy into the environment by human activities in amounts which are harmful to human health or other organisms.
- b) i) **Causes of Water Pollution**
- Poisonous chemicals like mercury, lead, and silver discharged into water bodies.
 - Heated water from industrial plants
 - Toxic chemicals used in agriculture such as insecticides, fungicides, nitrogenous and phosphorous fertilizers
 - Untreated sewage and domestic refuse discharged into water.
 - Oil from tankers and ship
 - Soil eroded by running water
- ii) **Effect to plants and animals**
- Toxic chemicals kill organisms, some are non-biodegradable therefore accumulate along food chain/undergo biomagnification.
 - Mercury limit transport of oxygen by occupying space occupied by hemoglobin in red blood cells
 - Heated water expels dissolved oxygen causing suffocation of aquatic organisms.
 - Nitrogenous wastes and sewage promote rapid growth of aquatic weeds which result into eutrophication
 - Sewage contains pathogenic micro-organisms like Escherichia coli which contaminate drinking water and cause animal diseases.
 - Oil floating on water reduce the amount of dissolved air which cause death.
 - Eroded soil make water turbid, hindering photosynthesis in aquatic weeds and unfit for consumption
- iii) **Methods of controlling water pollution.**
- Recycling industrial wastes before releasing into water ways.
 - Use of biological methods instead of chemical methods.
 - Encourage the use unleaded petrol
 - Public should be educated on correct amounts of organic fertilizers and pesticides to be used.
 - Building of gabions, terraces, mulching and growing of soil cover crops to reduce soil erosion.
 - Impose fines on ships and industries discharging oil and chemicals into water.
 - Proper treatment and disposal of sewage. There should be separate systems for disposal of sewage and drinking water

MARKING SCHEME
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BIOLOGY PAPER 3
231/3
BIOLOGY
PAPER3 (PRACTICAL)
MARKING SCHEME

1. a) (i) leaf
(ii) P – stoma; rej stomata.
Q – palisade cells/layer/mesophylls
R – Spongy layer/cell/mesophylls
S – vascular bundle/xylem/phloem
T – Intercellular/air spaces
U – Upper epidermis.
- b. (i) Site for photosynthesis.
(ii) Adaptation
- Numerous chloroplasts (which contain chlorophyll) for photosynthesis.
- Closely packed/vertical arrangement for maximum absorption of light.
- Positioned on the upper surface to absorb maximum light.
- c. - Light
- Temperature
- d. Length = 1cm accept ± 0.1
Actual length = $\frac{\text{length of image}}{\text{Magnification}}$
= $\frac{10,000}{5,000}$
= 2 μm $\rightarrow$ 2 micrometers .rej.if no units

2.
a) i) M :- Lumbar vertebra;
Rej.Lumbar alone/lumbar bone/lumbar vertebrae.
N: - Cervical vertebra;
Rej.Cervical bone/cervical alone/cervical vertebrae.
- ii) J: - Neural spine;
K: - Neural canal;
L: - Vertebrarterial canal;

b. i)

M	N
Broad /wide/large neural spine	Narrow/small neural spine;
Vertebrarterial canal absent	Vertebrarterial canal present;
Narrow neural canal	Wider neural canal
Large transverse processes	Small/shorter transverse precesses
Undivided/unwinged transverse processes	Divided/winged transverse processes

Metapophysis and anapophysis are present	Metapophysis and anapophysis absent.
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- ii) M:-abdomen/abdominal region;
N:- neck region

- c. Branched /winged/divided transverse processes to increase the surface area for muscles attachment;
Neural canal for passage of spinal cord;
Neural spine to increase the surface area for muscle attachment;
Vertebral canals for passage of blood vessels and nerves;
Neural arch and Centrum for protection of the spinal cord;

3 a)

Food substance	Procedure	observation	conclusion
starch	Put solution K in a test tube add iodine solution	Blue black	Presence of starch
	Put solution L in a test tube add iodine solution	No colour change/colour of iodine/brown	Absence of starch
Reducing sugars	Put solution K in a test tube and equal volume of benedicts solution; heat to boil	No colour change; colour of benedicts solution remains	Absence of reducing sugars
	Put solution L in a test tube Add benedict's solution; heat to boil.	Colour changes to green,yellow,orange ,brown	Presence of reducing sugars.

12/2 = 6marks

b.i)

Test with	procedure	observation	conclusion
Iodine solution	Put contents of visking tubing in a test tube add iodine solution	Blue black	Presence of starch
Benedicts solution	Put contents of visking tubing in a test tube add Benedict's solution heat	Colour changes from blue to green, yellow, orange brown	Presence of reducing sugars.

ii)

Test with	procedure	observation	conclusion
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Iodine solution	Put contents of the beaker in a test tube add iodine solution	No colour change/colour of iodine/brown.	Absence of starch.
Benedicts solution	Put contents of the beaker in a test tube add Benedict's solution Then heat	Colour changes from blue to green, yellow orange then brown.	Presence of reducing sugars

6 /2= 3 marks

N:B Mark procedure once. But must be correct at all cases.

- c. Reducing sugar molecules are small in sizes; able to pass through the semi – permeable membrane into the visking tubing by diffusion.

Starch molecules are too large; to pass through the semi -permeable membrane/visking tubing into the beaker;

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